

Evaluation Design and Model Settings

Evaluation design

Step	Implementation	Test data used?
Patient splits	50 fixed stratified holdout splits; 60 training and 26 test subjects per split	Only to define evaluation sets
Topology imputation	Training-set median imputation	No
Feature selection	Training-fold ANOVA-F ranking; each model selects K using 3 repeats of stratified 10-fold CV	No
Mask construction	WB/DMN preserve topology priors; SMN appends the NBS-weighted score	No
Scaling	StandardScaler fitted only on the corresponding training data	No
Model fitting	Five classifiers trained separately using their own selected K	No
Evaluation	Metrics computed in 26 test subjects and summarized across 50 splits	Yes

Fixed model parameters

Model	Fixed parameters
LinearSVC	class_weight=balanced; task-specific fixed C; max_iter=20000
LDA	solver=lsqr; shrinkage=auto
RF	n_estimators=500; max_features=sqrt; min_samples_leaf=2; class_weight=balanced
DT	max_depth=4; min_samples_leaf=2; class_weight=balanced
XGBoost	n_estimators=300; max_depth=3; learning_rate=0.03; subsample=0.8; colsample_bytree=0.8

Overall model comparison

Among all models, LinearSVC was identified as the most suitable model (mean AUC: 0.834, range: 0.777-0.873; Fig. 3A), followed by LDA (mean AUC: 0.796, range: 0.691-0.847), RF (mean AUC: 0.643, range: 0.393-0.856), XGBoost (mean AUC: 0.613, range: 0.430-0.799), and DT (mean AUC: 0.553, range: 0.388-0.682).

All models used the same outer patient splits and the same K-selection rule, while selecting K independently within the training data. Outer test subjects were not used for preprocessing or K selection.

LinearSVC: Detailed Results

Across the 18 actually trained tasks, LinearSVC achieved a mean AUC of 0.834 (range 0.777-0.873); 16/18 tasks reached AUC >= 0.80. It was the highest-performing and most stable model overall.

ROC-AUC Overview (mean +/- SD across 50 splits)

Network	FC no-mask	FC mask	Topology no-mask	Topology mask	FC+Topology no-mask	FC+Topology mask
WB	0.873 +/- 0.051	0.873 +/- 0.051	0.840 +/- 0.062	0.846 +/- 0.070	0.851 +/- 0.060	0.868 +/- 0.049
DMN	0.827 +/- 0.045	0.827 +/- 0.045	0.813 +/- 0.088	0.827 +/- 0.095	0.845 +/- 0.056	0.841 +/- 0.076
SMN	0.843 +/- 0.050	0.846 +/- 0.055	0.777 +/- 0.105	0.777 +/- 0.105	0.823 +/- 0.076	0.861 +/- 0.052
VN	0.794 +/- 0.068	0.794 +/- 0.068	0.800 +/- 0.068	0.800 +/- 0.068	0.838 +/- 0.060	0.838 +/- 0.060

Original: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.873 +/- 0.051	0.749 +/- 0.086	0.779 +/- 0.057	0.724 +/- 0.120	0.833 +/- 0.069	0.812 +/- 0.051	0.601 +/- 0.084
WB	Topology	0.840 +/- 0.062	0.600 +/- 0.115	0.755 +/- 0.083	0.744 +/- 0.167	0.766 +/- 0.102	0.762 +/- 0.078	0.550 +/- 0.106
WB	FC+Topology	0.851 +/- 0.060	0.664 +/- 0.136	0.756 +/- 0.063	0.680 +/- 0.128	0.831 +/- 0.074	0.802 +/- 0.057	0.573 +/- 0.090
DMN	FC	0.827 +/- 0.045	0.678 +/- 0.078	0.692 +/- 0.074	0.532 +/- 0.119	0.851 +/- 0.105	0.790 +/- 0.085	0.506 +/- 0.112
DMN	Topology	0.813 +/- 0.088	0.574 +/- 0.141	0.715 +/- 0.111	0.600 +/- 0.252	0.830 +/- 0.096	0.785 +/- 0.069	0.506 +/- 0.148
DMN	FC+Topology	0.845 +/- 0.056	0.617 +/- 0.088	0.730 +/- 0.065	0.736 +/- 0.137	0.724 +/- 0.105	0.726 +/- 0.077	0.512 +/- 0.081
SMN	FC	0.843 +/- 0.050	0.580 +/- 0.107	0.769 +/- 0.068	0.836 +/- 0.119	0.703 +/- 0.089	0.728 +/- 0.072	0.547 +/- 0.078
SMN	Topology	0.777 +/- 0.105	0.496 +/- 0.149	0.712 +/- 0.106	0.668 +/- 0.196	0.755 +/- 0.115	0.738 +/- 0.095	0.501 +/- 0.135
SMN	FC+Topology	0.823 +/- 0.076	0.560 +/- 0.127	0.770 +/- 0.089	0.816 +/- 0.161	0.724 +/- 0.105	0.742 +/- 0.086	0.555 +/- 0.112
VN	FC	0.794 +/- 0.068	0.582 +/- 0.136	0.702 +/- 0.088	0.576 +/- 0.174	0.828 +/- 0.083	0.779 +/- 0.068	0.499 +/- 0.126
VN	Topology	0.800 +/- 0.068	0.506 +/- 0.112	0.670 +/- 0.086	0.520 +/- 0.167	0.821 +/- 0.063	0.763 +/- 0.057	0.455 +/- 0.122
VN	FC+Topology	0.838 +/- 0.060	0.597 +/- 0.151	0.734 +/- 0.078	0.624 +/- 0.155	0.844 +/- 0.067	0.802 +/- 0.057	0.547 +/- 0.116

Mask: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.873 +/- 0.051	0.749 +/- 0.086	0.779 +/- 0.057	0.724 +/- 0.120	0.833 +/- 0.069	0.812 +/- 0.051	0.601 +/- 0.084
WB	Topology	0.846 +/- 0.070	0.631 +/- 0.125	0.751 +/- 0.077	0.812 +/- 0.124	0.690 +/- 0.148	0.713 +/- 0.113	0.537 +/- 0.104
WB	FC+Topology	0.868 +/- 0.049	0.660 +/- 0.136	0.775 +/- 0.048	0.828 +/- 0.090	0.722 +/- 0.071	0.742 +/- 0.055	0.556 +/- 0.060
DMN	FC	0.827 +/- 0.045	0.678 +/- 0.078	0.692 +/- 0.074	0.532 +/- 0.119	0.851 +/- 0.105	0.790 +/- 0.085	0.506 +/- 0.112
DMN	Topology	0.827 +/- 0.095	0.589 +/- 0.139	0.742 +/- 0.120	0.648 +/- 0.264	0.836 +/- 0.107	0.800 +/- 0.080	0.547 +/- 0.162
DMN	FC+Topology	0.841 +/- 0.076	0.641 +/- 0.107	0.756 +/- 0.064	0.680 +/- 0.151	0.831 +/- 0.079	0.802 +/- 0.054	0.570 +/- 0.085
SMN	FC	0.846 +/- 0.055	0.571 +/- 0.112	0.773 +/- 0.068	0.860 +/- 0.101	0.686 +/- 0.091	0.719 +/- 0.076	0.547 +/- 0.081
SMN	Topology	0.777 +/- 0.105	0.496 +/- 0.149	0.712 +/- 0.106	0.668 +/- 0.196	0.755 +/- 0.115	0.738 +/- 0.095	0.501 +/- 0.135
SMN	FC+Topology	0.861 +/- 0.052	0.588 +/- 0.121	0.796 +/- 0.058	0.900 +/- 0.116	0.691 +/- 0.083	0.732 +/- 0.063	0.567 +/- 0.066
VN	FC	0.794 +/- 0.068	0.582 +/- 0.136	0.702 +/- 0.088	0.576 +/- 0.174	0.828 +/- 0.083	0.779 +/- 0.068	0.499 +/- 0.126
VN	Topology	0.800 +/- 0.068	0.506 +/- 0.112	0.670 +/- 0.086	0.520 +/- 0.167	0.821 +/- 0.063	0.763 +/- 0.057	0.455 +/- 0.122
VN	FC+Topology	0.838 +/- 0.060	0.597 +/- 0.151	0.734 +/- 0.078	0.624 +/- 0.155	0.844 +/- 0.067	0.802 +/- 0.057	0.547 +/- 0.116

Where no Mask rule was defined, the corresponding Original result is displayed for completeness; it was not retrained and was excluded from the 18-task overall model summary.

LDA: Detailed Results

Across the 18 actually trained tasks, LDA achieved a mean AUC of 0.796 (range 0.691-0.847); 10/18 tasks reached AUC >= 0.80. It ranked second overall, with relatively strong FC-related performance but weaker topology-only performance in some subnetworks.

ROC-AUC Overview (mean +/- SD across 50 splits)

Network	FC no-mask	FC mask	Topology no-mask	Topology mask	FC+Topology no-mask	FC+Topology mask
WB	0.845 +/- 0.069	0.845 +/- 0.069	0.784 +/- 0.067	0.786 +/- 0.065	0.832 +/- 0.068	0.847 +/- 0.053
DMN	0.795 +/- 0.042	0.795 +/- 0.042	0.694 +/- 0.092	0.691 +/- 0.094	0.828 +/- 0.061	0.808 +/- 0.072
SMN	0.842 +/- 0.052	0.830 +/- 0.055	0.747 +/- 0.095	0.747 +/- 0.095	0.804 +/- 0.111	0.831 +/- 0.051
VN	0.792 +/- 0.077	0.792 +/- 0.077	0.774 +/- 0.062	0.774 +/- 0.062	0.806 +/- 0.057	0.806 +/- 0.057

Original: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.845 +/- 0.069	0.703 +/- 0.118	0.698 +/- 0.082	0.416 +/- 0.161	0.980 +/- 0.033	0.872 +/- 0.041	0.542 +/- 0.159
WB	Topology	0.784 +/- 0.067	0.493 +/- 0.111	0.574 +/- 0.076	0.216 +/- 0.171	0.932 +/- 0.053	0.795 +/- 0.039	0.262 +/- 0.179
WB	FC+Topology	0.832 +/- 0.068	0.646 +/- 0.139	0.674 +/- 0.083	0.376 +/- 0.165	0.972 +/- 0.029	0.858 +/- 0.039	0.490 +/- 0.158
DMN	FC	0.795 +/- 0.042	0.632 +/- 0.074	0.635 +/- 0.065	0.292 +/- 0.147	0.977 +/- 0.034	0.845 +/- 0.025	0.404 +/- 0.129
DMN	Topology	0.694 +/- 0.092	0.467 +/- 0.116	0.589 +/- 0.073	0.236 +/- 0.126	0.943 +/- 0.042	0.807 +/- 0.048	0.317 +/- 0.162
DMN	FC+Topology	0.828 +/- 0.061	0.617 +/- 0.103	0.665 +/- 0.081	0.380 +/- 0.163	0.950 +/- 0.050	0.841 +/- 0.047	0.469 +/- 0.159
SMN	FC	0.842 +/- 0.052	0.552 +/- 0.096	0.538 +/- 0.050	0.084 +/- 0.100	0.992 +/- 0.018	0.818 +/- 0.023	0.136 +/- 0.162
SMN	Topology	0.747 +/- 0.095	0.442 +/- 0.118	0.547 +/- 0.077	0.128 +/- 0.160	0.967 +/- 0.039	0.805 +/- 0.038	0.171 +/- 0.206
SMN	FC+Topology	0.804 +/- 0.111	0.529 +/- 0.154	0.530 +/- 0.049	0.084 +/- 0.108	0.976 +/- 0.038	0.805 +/- 0.029	0.122 +/- 0.155
VN	FC	0.792 +/- 0.077	0.599 +/- 0.137	0.617 +/- 0.070	0.264 +/- 0.143	0.970 +/- 0.038	0.835 +/- 0.037	0.363 +/- 0.180
VN	Topology	0.774 +/- 0.062	0.483 +/- 0.099	0.562 +/- 0.075	0.164 +/- 0.160	0.959 +/- 0.030	0.806 +/- 0.030	0.215 +/- 0.200
VN	FC+Topology	0.806 +/- 0.057	0.558 +/- 0.115	0.595 +/- 0.073	0.248 +/- 0.125	0.942 +/- 0.046	0.808 +/- 0.051	0.330 +/- 0.168

Mask: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.845 +/- 0.069	0.703 +/- 0.118	0.698 +/- 0.082	0.416 +/- 0.161	0.980 +/- 0.033	0.872 +/- 0.041	0.542 +/- 0.159
WB	Topology	0.786 +/- 0.065	0.495 +/- 0.111	0.569 +/- 0.074	0.204 +/- 0.169	0.933 +/- 0.055	0.793 +/- 0.040	0.247 +/- 0.182
WB	FC+Topology	0.847 +/- 0.053	0.668 +/- 0.111	0.663 +/- 0.077	0.352 +/- 0.159	0.973 +/- 0.034	0.854 +/- 0.036	0.466 +/- 0.147
DMN	FC	0.795 +/- 0.042	0.632 +/- 0.074	0.635 +/- 0.065	0.292 +/- 0.147	0.977 +/- 0.034	0.845 +/- 0.025	0.404 +/- 0.129
DMN	Topology	0.691 +/- 0.094	0.467 +/- 0.117	0.599 +/- 0.077	0.256 +/- 0.134	0.943 +/- 0.041	0.811 +/- 0.048	0.337 +/- 0.166
DMN	FC+Topology	0.808 +/- 0.072	0.611 +/- 0.107	0.658 +/- 0.074	0.348 +/- 0.150	0.969 +/- 0.040	0.849 +/- 0.039	0.458 +/- 0.150
SMN	FC	0.830 +/- 0.055	0.533 +/- 0.098	0.541 +/- 0.051	0.088 +/- 0.100	0.994 +/- 0.016	0.820 +/- 0.024	0.145 +/- 0.165
SMN	Topology	0.747 +/- 0.095	0.442 +/- 0.118	0.547 +/- 0.077	0.128 +/- 0.160	0.967 +/- 0.039	0.805 +/- 0.038	0.171 +/- 0.206
SMN	FC+Topology	0.831 +/- 0.051	0.523 +/- 0.097	0.529 +/- 0.045	0.076 +/- 0.098	0.983 +/- 0.025	0.808 +/- 0.021	0.115 +/- 0.150
VN	FC	0.792 +/- 0.077	0.599 +/- 0.137	0.617 +/- 0.070	0.264 +/- 0.143	0.970 +/- 0.038	0.835 +/- 0.037	0.363 +/- 0.180
VN	Topology	0.774 +/- 0.062	0.483 +/- 0.099	0.562 +/- 0.075	0.164 +/- 0.160	0.959 +/- 0.030	0.806 +/- 0.030	0.215 +/- 0.200
VN	FC+Topology	0.806 +/- 0.057	0.558 +/- 0.115	0.595 +/- 0.073	0.248 +/- 0.125	0.942 +/- 0.046	0.808 +/- 0.051	0.330 +/- 0.168

Where no Mask rule was defined, the corresponding Original result is displayed for completeness; it was not retrained and was excluded from the 18-task overall model summary.

RF: Detailed Results

Across the 18 actually trained tasks, RF achieved a mean AUC of 0.643 (range 0.393-0.856); 3/18 tasks reached AUC >= 0.80. It ranked third overall; several WB tasks were competitive, whereas performance varied across subnetworks and SMN Mask tasks.

ROC-AUC Overview (mean +/- SD across 50 splits)

Network	FC no-mask	FC mask	Topology no-mask	Topology mask	FC+Topology no-mask	FC+Topology mask
WB	0.844 +/- 0.066	0.844 +/- 0.066	0.687 +/- 0.078	0.687 +/- 0.079	0.850 +/- 0.058	0.856 +/- 0.054
DMN	0.622 +/- 0.155	0.622 +/- 0.155	0.695 +/- 0.112	0.707 +/- 0.114	0.612 +/- 0.133	0.664 +/- 0.098
SMN	0.498 +/- 0.141	0.408 +/- 0.144	0.591 +/- 0.139	0.591 +/- 0.139	0.549 +/- 0.098	0.393 +/- 0.122
VN	0.561 +/- 0.091	0.561 +/- 0.091	0.662 +/- 0.066	0.662 +/- 0.066	0.682 +/- 0.063	0.682 +/- 0.063

Original: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.844 +/- 0.066	0.657 +/- 0.170	0.599 +/- 0.097	0.220 +/- 0.177	0.977 +/- 0.039	0.832 +/- 0.054	0.316 +/- 0.236
WB	Topology	0.687 +/- 0.078	0.419 +/- 0.110	0.508 +/- 0.030	0.024 +/- 0.066	0.992 +/- 0.020	0.806 +/- 0.015	0.036 +/- 0.100
WB	FC+Topology	0.850 +/- 0.058	0.648 +/- 0.171	0.603 +/- 0.101	0.228 +/- 0.185	0.977 +/- 0.039	0.833 +/- 0.055	0.325 +/- 0.240
DMN	FC	0.622 +/- 0.155	0.381 +/- 0.169	0.522 +/- 0.063	0.064 +/- 0.117	0.981 +/- 0.032	0.805 +/- 0.037	0.099 +/- 0.175
DMN	Topology	0.695 +/- 0.112	0.479 +/- 0.168	0.605 +/- 0.120	0.328 +/- 0.264	0.883 +/- 0.078	0.776 +/- 0.061	0.323 +/- 0.225
DMN	FC+Topology	0.612 +/- 0.133	0.358 +/- 0.144	0.500 +/- 0.027	0.012 +/- 0.048	0.989 +/- 0.021	0.801 +/- 0.020	0.020 +/- 0.080
SMN	FC	0.498 +/- 0.141	0.295 +/- 0.100	0.516 +/- 0.083	0.220 +/- 0.141	0.812 +/- 0.063	0.698 +/- 0.062	0.217 +/- 0.134
SMN	Topology	0.591 +/- 0.139	0.371 +/- 0.109	0.581 +/- 0.090	0.300 +/- 0.173	0.862 +/- 0.076	0.754 +/- 0.066	0.308 +/- 0.168
SMN	FC+Topology	0.549 +/- 0.098	0.337 +/- 0.092	0.545 +/- 0.067	0.220 +/- 0.116	0.870 +/- 0.061	0.745 +/- 0.055	0.248 +/- 0.110
VN	FC	0.561 +/- 0.091	0.320 +/- 0.115	0.519 +/- 0.089	0.212 +/- 0.159	0.826 +/- 0.104	0.708 +/- 0.085	0.214 +/- 0.153
VN	Topology	0.662 +/- 0.066	0.409 +/- 0.108	0.546 +/- 0.065	0.168 +/- 0.124	0.925 +/- 0.050	0.779 +/- 0.045	0.217 +/- 0.156
VN	FC+Topology	0.682 +/- 0.063	0.466 +/- 0.100	0.606 +/- 0.059	0.284 +/- 0.122	0.929 +/- 0.060	0.805 +/- 0.047	0.351 +/- 0.137

Mask: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.844 +/- 0.066	0.657 +/- 0.170	0.599 +/- 0.097	0.220 +/- 0.177	0.977 +/- 0.039	0.832 +/- 0.054	0.316 +/- 0.236
WB	Topology	0.687 +/- 0.079	0.419 +/- 0.110	0.510 +/- 0.032	0.028 +/- 0.070	0.992 +/- 0.020	0.807 +/- 0.016	0.043 +/- 0.109
WB	FC+Topology	0.856 +/- 0.054	0.669 +/- 0.148	0.604 +/- 0.082	0.228 +/- 0.157	0.981 +/- 0.037	0.836 +/- 0.045	0.328 +/- 0.215
DMN	FC	0.622 +/- 0.155	0.381 +/- 0.169	0.522 +/- 0.063	0.064 +/- 0.117	0.981 +/- 0.032	0.805 +/- 0.037	0.099 +/- 0.175
DMN	Topology	0.707 +/- 0.114	0.503 +/- 0.175	0.627 +/- 0.125	0.372 +/- 0.277	0.883 +/- 0.076	0.785 +/- 0.060	0.361 +/- 0.232
DMN	FC+Topology	0.664 +/- 0.098	0.402 +/- 0.127	0.525 +/- 0.053	0.072 +/- 0.097	0.977 +/- 0.028	0.803 +/- 0.032	0.115 +/- 0.155
SMN	FC	0.408 +/- 0.144	0.265 +/- 0.109	0.508 +/- 0.067	0.260 +/- 0.101	0.755 +/- 0.081	0.660 +/- 0.070	0.229 +/- 0.084
SMN	Topology	0.591 +/- 0.139	0.371 +/- 0.109	0.581 +/- 0.090	0.300 +/- 0.173	0.862 +/- 0.076	0.754 +/- 0.066	0.308 +/- 0.168
SMN	FC+Topology	0.393 +/- 0.122	0.288 +/- 0.116	0.523 +/- 0.066	0.236 +/- 0.088	0.810 +/- 0.070	0.700 +/- 0.065	0.238 +/- 0.098
VN	FC	0.561 +/- 0.091	0.320 +/- 0.115	0.519 +/- 0.089	0.212 +/- 0.159	0.826 +/- 0.104	0.708 +/- 0.085	0.214 +/- 0.153
VN	Topology	0.662 +/- 0.066	0.409 +/- 0.108	0.546 +/- 0.065	0.168 +/- 0.124	0.925 +/- 0.050	0.779 +/- 0.045	0.217 +/- 0.156
VN	FC+Topology	0.682 +/- 0.063	0.466 +/- 0.100	0.606 +/- 0.059	0.284 +/- 0.122	0.929 +/- 0.060	0.805 +/- 0.047	0.351 +/- 0.137

Where no Mask rule was defined, the corresponding Original result is displayed for completeness; it was not retrained and was excluded from the 18-task overall model summary.

DT: Detailed Results

Across the 18 actually trained tasks, DT achieved a mean AUC of 0.553 (range 0.388-0.682); 0/18 tasks reached AUC >= 0.80. It had the lowest overall AUC and showed substantial variation across task combinations.

ROC-AUC Overview (mean +/- SD across 50 splits)

Network	FC no-mask	FC mask	Topology no-mask	Topology mask	FC+Topology no-mask	FC+Topology mask
WB	0.682 +/- 0.096	0.682 +/- 0.096	0.562 +/- 0.087	0.562 +/- 0.086	0.673 +/- 0.100	0.664 +/- 0.096
DMN	0.647 +/- 0.071	0.647 +/- 0.071	0.552 +/- 0.123	0.567 +/- 0.122	0.521 +/- 0.092	0.577 +/- 0.070
SMN	0.388 +/- 0.121	0.429 +/- 0.176	0.546 +/- 0.118	0.546 +/- 0.118	0.460 +/- 0.112	0.477 +/- 0.142
VN	0.492 +/- 0.085	0.492 +/- 0.085	0.562 +/- 0.083	0.562 +/- 0.083	0.592 +/- 0.107	0.592 +/- 0.107

Original: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.682 +/- 0.096	0.408 +/- 0.137	0.679 +/- 0.095	0.448 +/- 0.188	0.910 +/- 0.070	0.821 +/- 0.062	0.485 +/- 0.156
WB	Topology	0.562 +/- 0.087	0.270 +/- 0.107	0.558 +/- 0.083	0.276 +/- 0.166	0.839 +/- 0.086	0.731 +/- 0.068	0.271 +/- 0.164
WB	FC+Topology	0.673 +/- 0.100	0.408 +/- 0.150	0.670 +/- 0.096	0.424 +/- 0.170	0.916 +/- 0.067	0.822 +/- 0.067	0.478 +/- 0.167
DMN	FC	0.647 +/- 0.071	0.300 +/- 0.063	0.637 +/- 0.082	0.464 +/- 0.159	0.810 +/- 0.069	0.743 +/- 0.060	0.406 +/- 0.128
DMN	Topology	0.552 +/- 0.123	0.263 +/- 0.097	0.563 +/- 0.116	0.412 +/- 0.223	0.713 +/- 0.127	0.655 +/- 0.102	0.310 +/- 0.172
DMN	FC+Topology	0.521 +/- 0.092	0.227 +/- 0.044	0.520 +/- 0.092	0.192 +/- 0.185	0.849 +/- 0.057	0.722 +/- 0.053	0.194 +/- 0.171
SMN	FC	0.388 +/- 0.121	0.216 +/- 0.047	0.433 +/- 0.141	0.136 +/- 0.215	0.730 +/- 0.215	0.616 +/- 0.173	0.107 +/- 0.164
SMN	Topology	0.546 +/- 0.118	0.265 +/- 0.081	0.554 +/- 0.075	0.328 +/- 0.184	0.780 +/- 0.093	0.693 +/- 0.062	0.275 +/- 0.136
SMN	FC+Topology	0.460 +/- 0.112	0.223 +/- 0.061	0.456 +/- 0.107	0.124 +/- 0.161	0.788 +/- 0.092	0.660 +/- 0.091	0.125 +/- 0.164
VN	FC	0.492 +/- 0.085	0.219 +/- 0.054	0.494 +/- 0.085	0.276 +/- 0.171	0.711 +/- 0.094	0.628 +/- 0.073	0.214 +/- 0.129
VN	Topology	0.562 +/- 0.083	0.283 +/- 0.079	0.555 +/- 0.078	0.296 +/- 0.199	0.813 +/- 0.088	0.714 +/- 0.054	0.261 +/- 0.151
VN	FC+Topology	0.592 +/- 0.107	0.287 +/- 0.094	0.589 +/- 0.102	0.356 +/- 0.195	0.823 +/- 0.097	0.733 +/- 0.082	0.334 +/- 0.150

Mask: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.682 +/- 0.096	0.408 +/- 0.137	0.679 +/- 0.095	0.448 +/- 0.188	0.910 +/- 0.070	0.821 +/- 0.062	0.485 +/- 0.156
WB	Topology	0.562 +/- 0.086	0.271 +/- 0.107	0.558 +/- 0.082	0.276 +/- 0.166	0.839 +/- 0.086	0.731 +/- 0.068	0.271 +/- 0.163
WB	FC+Topology	0.664 +/- 0.096	0.400 +/- 0.141	0.662 +/- 0.096	0.396 +/- 0.174	0.928 +/- 0.068	0.825 +/- 0.066	0.465 +/- 0.177
DMN	FC	0.647 +/- 0.071	0.300 +/- 0.063	0.637 +/- 0.082	0.464 +/- 0.159	0.810 +/- 0.069	0.743 +/- 0.060	0.406 +/- 0.128
DMN	Topology	0.567 +/- 0.122	0.267 +/- 0.094	0.573 +/- 0.114	0.436 +/- 0.209	0.710 +/- 0.128	0.657 +/- 0.105	0.329 +/- 0.160
DMN	FC+Topology	0.577 +/- 0.070	0.246 +/- 0.062	0.577 +/- 0.070	0.304 +/- 0.147	0.850 +/- 0.055	0.745 +/- 0.044	0.305 +/- 0.120
SMN	FC	0.429 +/- 0.176	0.239 +/- 0.072	0.436 +/- 0.130	0.160 +/- 0.190	0.711 +/- 0.176	0.605 +/- 0.147	0.132 +/- 0.148
SMN	Topology	0.546 +/- 0.118	0.265 +/- 0.081	0.554 +/- 0.075	0.328 +/- 0.184	0.780 +/- 0.093	0.693 +/- 0.062	0.275 +/- 0.136
SMN	FC+Topology	0.477 +/- 0.142	0.248 +/- 0.079	0.456 +/- 0.126	0.200 +/- 0.221	0.712 +/- 0.177	0.614 +/- 0.140	0.152 +/- 0.159
VN	FC	0.492 +/- 0.085	0.219 +/- 0.054	0.494 +/- 0.085	0.276 +/- 0.171	0.711 +/- 0.094	0.628 +/- 0.073	0.214 +/- 0.129
VN	Topology	0.562 +/- 0.083	0.283 +/- 0.079	0.555 +/- 0.078	0.296 +/- 0.199	0.813 +/- 0.088	0.714 +/- 0.054	0.261 +/- 0.151
VN	FC+Topology	0.592 +/- 0.107	0.287 +/- 0.094	0.589 +/- 0.102	0.356 +/- 0.195	0.823 +/- 0.097	0.733 +/- 0.082	0.334 +/- 0.150

Where no Mask rule was defined, the corresponding Original result is displayed for completeness; it was not retrained and was excluded from the 18-task overall model summary.

XGBoost: Detailed Results

Across the 18 actually trained tasks, XGBoost achieved a mean AUC of 0.613 (range 0.430-0.799); 0/18 tasks reached AUC >= 0.80. It ranked fourth overall and approached an AUC of 0.80 in its best task, but performance was lower in several subnetwork tasks.

ROC-AUC Overview (mean +/- SD across 50 splits)

Network	FC no-mask	FC mask	Topology no-mask	Topology mask	FC+Topology no-mask	FC+Topology mask
WB	0.791 +/- 0.054	0.791 +/- 0.054	0.669 +/- 0.073	0.670 +/- 0.075	0.794 +/- 0.050	0.799 +/- 0.051
DMN	0.542 +/- 0.131	0.542 +/- 0.131	0.654 +/- 0.075	0.665 +/- 0.079	0.538 +/- 0.114	0.564 +/- 0.101
SMN	0.514 +/- 0.152	0.430 +/- 0.149	0.586 +/- 0.172	0.586 +/- 0.172	0.503 +/- 0.117	0.446 +/- 0.146
VN	0.566 +/- 0.094	0.566 +/- 0.094	0.645 +/- 0.082	0.645 +/- 0.082	0.668 +/- 0.084	0.668 +/- 0.084

Original: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.791 +/- 0.054	0.665 +/- 0.070	0.691 +/- 0.064	0.392 +/- 0.121	0.990 +/- 0.024	0.875 +/- 0.034	0.539 +/- 0.136
WB	Topology	0.669 +/- 0.073	0.427 +/- 0.118	0.523 +/- 0.052	0.080 +/- 0.107	0.966 +/- 0.033	0.795 +/- 0.030	0.115 +/- 0.151
WB	FC+Topology	0.794 +/- 0.050	0.672 +/- 0.068	0.694 +/- 0.069	0.396 +/- 0.131	0.991 +/- 0.023	0.877 +/- 0.035	0.544 +/- 0.145
DMN	FC	0.542 +/- 0.131	0.320 +/- 0.138	0.529 +/- 0.067	0.084 +/- 0.135	0.974 +/- 0.026	0.803 +/- 0.032	0.120 +/- 0.184
DMN	Topology	0.654 +/- 0.075	0.401 +/- 0.127	0.591 +/- 0.109	0.288 +/- 0.226	0.894 +/- 0.058	0.778 +/- 0.055	0.304 +/- 0.216
DMN	FC+Topology	0.538 +/- 0.114	0.300 +/- 0.101	0.492 +/- 0.047	0.032 +/- 0.074	0.952 +/- 0.047	0.775 +/- 0.043	0.049 +/- 0.115
SMN	FC	0.514 +/- 0.152	0.297 +/- 0.087	0.551 +/- 0.088	0.240 +/- 0.190	0.862 +/- 0.065	0.742 +/- 0.051	0.244 +/- 0.170
SMN	Topology	0.586 +/- 0.172	0.403 +/- 0.149	0.558 +/- 0.091	0.228 +/- 0.157	0.889 +/- 0.081	0.762 +/- 0.074	0.267 +/- 0.172
SMN	FC+Topology	0.503 +/- 0.117	0.331 +/- 0.102	0.540 +/- 0.075	0.200 +/- 0.140	0.881 +/- 0.065	0.750 +/- 0.058	0.226 +/- 0.146
VN	FC	0.566 +/- 0.094	0.351 +/- 0.111	0.550 +/- 0.073	0.200 +/- 0.134	0.899 +/- 0.064	0.765 +/- 0.057	0.239 +/- 0.154
VN	Topology	0.645 +/- 0.082	0.426 +/- 0.125	0.565 +/- 0.088	0.192 +/- 0.156	0.937 +/- 0.053	0.794 +/- 0.058	0.256 +/- 0.195
VN	FC+Topology	0.668 +/- 0.084	0.449 +/- 0.144	0.561 +/- 0.081	0.180 +/- 0.163	0.942 +/- 0.044	0.795 +/- 0.044	0.230 +/- 0.200

Mask: Detailed Metrics (mean +/- SD across 50 splits)

Network	Features	ROC-AUC	PR-AUC	Balanced Accuracy	Sensitivity	Specificity	Accuracy	F1
WB	FC	0.791 +/- 0.054	0.665 +/- 0.070	0.691 +/- 0.064	0.392 +/- 0.121	0.990 +/- 0.024	0.875 +/- 0.034	0.539 +/- 0.136
WB	Topology	0.670 +/- 0.075	0.425 +/- 0.118	0.524 +/- 0.051	0.084 +/- 0.108	0.964 +/- 0.038	0.795 +/- 0.031	0.119 +/- 0.151
WB	FC+Topology	0.799 +/- 0.051	0.670 +/- 0.080	0.681 +/- 0.063	0.368 +/- 0.124	0.994 +/- 0.016	0.874 +/- 0.028	0.519 +/- 0.137
DMN	FC	0.542 +/- 0.131	0.320 +/- 0.138	0.529 +/- 0.067	0.084 +/- 0.135	0.974 +/- 0.026	0.803 +/- 0.032	0.120 +/- 0.184
DMN	Topology	0.665 +/- 0.079	0.431 +/- 0.143	0.609 +/- 0.115	0.320 +/- 0.242	0.897 +/- 0.056	0.786 +/- 0.053	0.331 +/- 0.229
DMN	FC+Topology	0.564 +/- 0.101	0.314 +/- 0.101	0.497 +/- 0.046	0.048 +/- 0.086	0.947 +/- 0.042	0.774 +/- 0.036	0.067 +/- 0.123
SMN	FC	0.430 +/- 0.149	0.263 +/- 0.096	0.526 +/- 0.071	0.196 +/- 0.143	0.855 +/- 0.066	0.728 +/- 0.054	0.207 +/- 0.136
SMN	Topology	0.586 +/- 0.172	0.403 +/- 0.149	0.558 +/- 0.091	0.228 +/- 0.157	0.889 +/- 0.081	0.762 +/- 0.074	0.267 +/- 0.172
SMN	FC+Topology	0.446 +/- 0.146	0.270 +/- 0.092	0.520 +/- 0.071	0.196 +/- 0.148	0.845 +/- 0.071	0.720 +/- 0.056	0.199 +/- 0.143
VN	FC	0.566 +/- 0.094	0.351 +/- 0.111	0.550 +/- 0.073	0.200 +/- 0.134	0.899 +/- 0.064	0.765 +/- 0.057	0.239 +/- 0.154
VN	Topology	0.645 +/- 0.082	0.426 +/- 0.125	0.565 +/- 0.088	0.192 +/- 0.156	0.937 +/- 0.053	0.794 +/- 0.058	0.256 +/- 0.195
VN	FC+Topology	0.668 +/- 0.084	0.449 +/- 0.144	0.561 +/- 0.081	0.180 +/- 0.163	0.942 +/- 0.044	0.795 +/- 0.044	0.230 +/- 0.200

Where no Mask rule was defined, the corresponding Original result is displayed for completeness; it was not retrained and was excluded from the 18-task overall model summary.